TELCONG NETPLAN

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Spectrum Analysis Exhibits for Form 7B

Annex D-4 to the Form NCC-7B Spectrum Application

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Scope	Spectrum analysis exhibits for 3500-3600 MHz application
Use	Form NCC-7B Section D, Annex D-4

Exhibit summary

This document presents 5 technical exhibits supporting the TelcoNG NetPlan application for assignment of 100 MHz in the 3500-3600 MHz sub-range for the Abuja CBD 5G Densification Pilot. Each exhibit addresses a specific aspect of the technical case the Commission's hearing panel is likely to probe.

Exhibit 1: RF measurement report for 4 candidate sites

TelcoNG NetPlan conducted in-band RF measurements at 4 candidate sites in the Abuja CBD during August 2026 to characterise the noise floor and incumbent activity in the proposed 3500-3600 MHz band. Measurements used calibrated spectrum analysis equipment (Rohde & Schwarz FSW-26 with calibrated omnidirectional antenna at 4 m height). Each site was measured over a 24-hour cycle to capture diurnal variation. Summary results below.

Candidate site	Location	Mean noise floor (dBm/MHz)	Max incumbent signal (dBm)	Incumbent identified
CBD-S01	Wuse, junction of Aguiyi Ironsi & Ahmadu Bello	-108	below noise floor	None
CBD-S02	Maitama, near Plot 274 Adetokunbo Ademola	-107	below noise floor	None
CBD-S03	Garki, ECOWAS secretariat corner	-109	below noise floor	None
CBD-S04	Central Area, near CBN HQ	-107	below noise floor	None

Conclusion: the 3500-3600 MHz sub-range is clear of incumbent activity in Abuja FCT, consistent with the NCC's published unassigned-band status.

Exhibit 2: Interference analysis vs incumbent 3.5 GHz users

Although there are no incumbent users in the 3500-3600 MHz sub-range in Abuja FCT, there are incumbent users in adjacent sub-ranges and in adjacent geographic areas (Lagos, Port Harcourt). This exhibit analyses the potential interference from the proposed pilot toward those incumbents.

Incumbent	Sub-range	Geographic area	Distance from CBD pilot edge	Predicted interference
MTN Nigeria	3400-3500 MHz	Lagos, Port Harcourt	~750 km, ~450 km	Negligible (below -130 dBm)
Mafab Communications	3500-3600 MHz	Lagos	~750 km	Negligible (different geographic area)
9mobile (pilot)	3700-3800 MHz	Lagos	~750 km	Negligible (different geographic area + frequency separation)
FSS earth stations	3800-4200 MHz	Multiple (national)	Variable	Within ITU-R M.2117 criteria

Exhibit 3: Coordination analysis with satellite earth stations

Four FSS earth stations operating in the 3800-4200 MHz adjacent band have been identified within 200 km of the Abuja CBD: two C-band stations in the Federal Capital Territory, one in Niger State, and one in Kogi State. Coordination has been initiated per Recommendation ITU-R M.2117. Results to date: three of four operators have provided written non-objection; one (Niger State station) is in active discussion with TelcoNG, with conclusion expected within the 60-day Commission determination period.

Exhibit 4: Propagation model verification

TelcoNG's coverage predictions for the Abuja CBD pilot use the Ericsson 9999 model calibrated against 2024-2025 measurement campaigns in Abuja, Lagos, and Port Harcourt. The calibration data set comprises 4,200 drive-test measurement points across the three cities. Model verification achieves an RMS error of 6.2 dB against measured data, within industry-standard acceptance (typically ≤8 dB RMS for mid-band urban propagation models).

Exhibit 5: Test transmission log

TelcoNG conducted a 3-day test transmission programme at the 4 candidate sites in September 2026 under a special test permit issued by the Commission. Test transmissions used 20 MHz bandwidth at the 3500-3520 MHz sub-band, with output power of 25 dBm at the antenna port. No interference complaints were received from adjacent operators or from the public during the test period. Test logs are available on request.